

Vol 4 Chapter 10 — Dark Energy and Dark Matter

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Dark energy (responsible for accelerated cosmic expansion) and dark matter (responsible for galaxy rotation curves and cosmic structure formation) together account for about 95% of the universe’s energy density. Standard physics has no derivation of either; both are empirical inputs in the Λ CDM model.

BST derives both from substrate structure.

10.1 Dark energy as substrate vacuum

Dark energy in BST is the cosmological constant Λ of Chapter 4 — the substrate’s Zone-2 Bergman vacuum integrated at cosmological scale. No additional dynamical dark-energy field is required; the substrate’s vacuum is what produces accelerated expansion.

10.2 Dark matter as substrate Shannon channel

Dark matter has no substrate-side particle counterpart. The Five-Absence Prediction Set (Volume 2 Chapter 11) rules out a sterile DM particle. Instead, dark matter signatures are substrate Shannon channels — substrate-internal information channels carrying gravitational coupling without producing electromagnetic interactions. Galaxy rotation curves and cosmic structure formation emerge from the substrate’s Shannon-channel gravitational coupling, with no need for unseen particles.

10.3 The dark-matter-to-baryon ratio

$$\frac{\Omega_{DM}}{\Omega_b} = \frac{3n_C + 1}{N_c} = \frac{16}{3} \approx 5.33.$$

Measured ratio (Planck): $\Omega_{DM}/\Omega_b \approx 5.36$. BST prediction matches at 0.58%. The structural reading uses $n_C = 5$ and $N_c = 3$ via the integer combination $(3n_C + 1)/N_c$ that emerges from the substrate’s Shannon-channel allocation at the cosmological scale.

10.4 No WIMP, no axion, no sterile neutrino

The framework’s prediction is that no dark matter particle exists. Experimental searches for WIMPs (direct-detection experiments LUX, XENON1T, etc.) and axions and sterile neutrinos will continue to find nothing. This is one of BST’s sharpest beyond-Standard-Model predictions.

10.5 What comes next

Chapter 11 — gravitational waves and NANOGrav.

Where to look this up: DM/baryon ratio: substrate Shannon channel framework + integer derivation. For dark-matter experimental status: Particle Data Group annual reviews. For dark-energy reviews: Frieman, Turner, Huterer 2008, *Annual Review of Astronomy and Astrophysics* 46:385.